

Zachary James

Louisiana (Willing to Relocate) | US Citizen | GitHub: [zach15james](https://github.com/zach15james) | LinkedIn: [zrjames](https://www.linkedin.com/in/zrjames) | zach15james@proton.me

EDUCATION

Louisiana State University

B.S. Computer Science

B.S. Mathematics

Psychology & Honors Minor

Baton Rouge, LA

GPA: 3.4

Awards: Honors College, Dean's List, President's Honor Roll, S&B Engineers Scholarship, ICPC Competitive Programming team, Startup LSU

Selected Coursework: Abstract Algebra II, Applied Optimization Theory, Numerical Linear Algebra, Operating Systems, Computer Networks, OOP, Malware Analysis, AI, Math Stats, Software Systems

EXPERIENCE

LSU Department of Mathematics

Baton Rouge, LA

Undergraduate Research Assistant (Python, Scikitlearn, PyTorch, Numpy)

May 2024 – March 2026

- NSF-Funded topological data analysis project where I designed, rebuilt, and optimized graphing algorithms, various embedding metrics, and provided prompt updates, runs, and input for Dr. Drenska
- Developed and applied machine learning (traditional regressions, neural nets, and a novel p-Laplacian semi-supervised regression algorithm) to predict health metrics (mass, body fat%, bone density); dealt w/ various library issues, implemented thread pools, and even implemented some methods from scratch to fit research needs; [<https://www.frontiersin.org/journals/bioinformatics/articles/10.3389/fbinf.2026.1722578/full>]

Brick Bridge Consulting

Louisville, KY (Hybrid)

Software Developer (Python, JS, PHP, HTML/CSS, Zapier)

December 2022 – January 2024

- Designed and independently delivered HRM, billing, financial dashboards, and complete workflows for onboarding, automating forms, &c for medical staffing firms (even before AI was cool)
- Built internal analytics and financial tools for client businesses, including invoice generation, rate-card calculators, tax logic, and reporting dashboards for executives to make data-driven decisions
- Developed multi-interface CRUD application for a medical-legal startup, including PHP API integrations, HTML/CSS frontend, and leveled access control (know how to build fast MVPs that scale)

LSU Division of Computer Science & Engineering

Baton Rouge, LA

Undergraduate Teaching Assistant (Java)

December 2022 – May 2024

- TA for 100+ student Java programming labs; grading, proctoring, one-on-one tutoring

PROJECTS

Thymos

- Building a zero-dependency compositional compute runtime for typed tensor graphs that validates, memory-plans, and executes via small backend kernels (C, multithreading, arena memory allocation, parallelization)

Obedient Jarvis

- Deployed a personal agent network utilizing my Jetson Nano for private, unrestricted local inference for research workloads routed to with a repurposed laptop server for email, calendar, scripting, & search tasks (Tailscale, Proton VPN+Mail, llama-server)

Class Projects

- Software Systems: [<https://github.com/zach15james/OpTraxx-public>] (React, Firebase, Vite, Tailwind, Claude)
- OOP in C++: MathDoku game (C++, CMake, CTest, Raylib) (no AI used)
- Database: rideshare app (sqlite C++ api, MySQL, HTML/CSS) (no AI used)

Malware Writeups (Ghidra, IDA Pro, ProcMon, gdb, Linux, Sockets API) (no AI-used)

- Writeups of reverse-engineering malware (Windows 'SQL Slammer', MSDOS 'Michelangelo', DOS7 .COM infector, socket beacon-to-botnet) [<https://resume.agapebloom.com/>], as well as these fun puzzles: github.com/zach15james/xorpd;

UVA (ICPC-style coding problems; C, C++) [<https://uhunt.onlinejudge.org/id/1641288>] (no AI-used)